

How Much of the Cross-Dataset Gap Do Dataset Artifacts Explain? A Controlled Study of Fake-News Detectors

[TODO: Your name]

[TODO: Department of Computer Engineering, College, City, Country]

[TODO: email]

Abstract—Fake-news detectors that score near-perfectly within one dataset often fail on another, and dataset artifacts such as publisher markers, document length and named entities are commonly blamed, but how much of the gap they explain is rarely measured. We audit three public English news corpora (ISOT, McIntire, WELFake) and find that WELFake contains every cleaned ISOT and McIntire document and that its label direction is documented inconsistently, so we use a non-overlapping WELFake residual as the third dataset. In a controlled factorial design over source-marker removal, length-matched resampling and entity masking, with token-matched random-deletion controls, equal-size class-balanced samples, group-aware splits and five seeds, applying all three interventions reduces the cross-dataset AUC gap of a TF-IDF logistic-regression detector from 0.212 to 0.157, leaving about three quarters unexplained. Length matching and entity masking each shrink the gap, whereas removing web boilerplate lowers transfer. Effects are strongly pair-specific: they help transfers involving ISOT and slightly hurt transfer between the two datasets built from similar sources. For fine-tuned DistilBERT the picture differs: masking gains match those of random token deletion, and entity masking is worse than random deletion. We discuss what these results do and do not show about artifacts as an explanation of generalization failure, and we provide the audit and experiment code.

Index Terms—fake news detection, cross-dataset generalization, shortcut learning, dataset artifacts, dataset audit, controlled experiment

I. INTRODUCTION

Automatic fake-news detectors are routinely reported with accuracies above 99% on public benchmarks, yet the same models often degrade sharply when evaluated on a different dataset [1]–[3]. A common explanation is that benchmarks contain artifacts: properties that correlate with the label in one collection but say little about veracity in another, for example publisher markers, document length, or the entities that happened to be in the news when the data were collected. This is an instance of shortcut learning [4], well documented for natural language inference [5]–[7] and increasingly for misinformation detection [8]–[11].

In the work we reviewed, what is missing is a quantitative answer to a simple question: of the performance drop between a model’s own dataset and another dataset, how much is attributable to each family of artifacts? Existing studies tend to examine one corpus [8], one artifact type [11], [12], or report transfer matrices without interventions [3]. **[TODO: After**

reading the closest papers in full, confirm and sharpen this positioning sentence.]

We make four contributions.

- 1) **A dataset audit.** We show that WELFake contains every cleaned ISOT and McIntire document, that ISOT carries about 13% exact duplicates, and that the WELFake label direction is described inconsistently; we resolve it empirically.
- 2) **A controlled attribution design.** A factorial design over source markers, length and entities, with token-matched random-deletion controls, equal-size class-balanced samples, group-aware splits and per-pair reporting.
- 3) **Findings for a linear model.** About a quarter of the TF-IDF cross-dataset AUC gap is explained; length and entity cues contribute, web boilerplate carries transferable signal, and effects differ by dataset pair.
- 4) **A model-dependence result.** For fine-tuned DistilBERT, masking gains are matched by random token deletion, which shows why such controls are necessary.

All claims are limited to three English news corpora from 2016–2017 and two model families.

II. PROBLEM STATEMENT

Let s and t be two datasets and f_s a classifier trained on s . We write $\text{AUC}_{\text{in}}(s)$ for its ROC-AUC on held-out documents of s and $\text{AUC}_{\text{x}}(s \rightarrow t)$ for its ROC-AUC on t . The cross-dataset gap is

$$G(s \rightarrow t) = \text{AUC}_{\text{in}}(s) - \text{AUC}_{\text{x}}(s \rightarrow t). \quad (1)$$

An intervention ϕ transforms every document of every dataset (and is applied identically to training and test text); G_ϕ is the gap measured on the transformed data. The share of the gap removed is $1 - \bar{G}_\phi / \bar{G}_0$, with $\bar{G}$ the mean over ordered pairs. The problem is to estimate how this share depends on $\phi \in \{\text{source markers, length, entities}\}$ and their combinations, *above what deleting the same amount of text at random would do*, while avoiding four threats: (i) overlapping documents between datasets, (ii) incorrect or inconsistent labels, (iii) near-duplicates that leak between training and test splits, and (iv) differences in class prior and sample size across conditions.

III. OBJECTIVES

- O1 Audit three public corpora for overlap, duplication and label direction, and construct a defensible set of datasets.
- O2 Quantify, for a TF-IDF logistic-regression detector, how much of the cross-dataset gap is removed by source-marker removal, length matching and entity masking, alone and combined.
- O3 Separate the specific effect of each intervention from the effect of deleting an equal number of random words.
- O4 Report how the effects vary across ordered dataset pairs, and whether they carry over to a fine-tuned transformer (DistilBERT).

IV. LITERATURE REVIEW

A. Cross-dataset generalization in fake-news detection

Surveys describe the field’s methods and data [13]–[15]. Several studies report that detectors trained on one dataset transfer poorly: cross-domain failures were documented for classical and neural models [1], generalizability was examined across datasets [2], and evaluation practice was shown to depend on dataset, protocol and metric [16], [17]. A recent experimental comparison trains twelve approaches on ten datasets under in-domain, multi-domain and cross-domain protocols, and treats each dataset as a domain [3]. These studies document the gap; they do not intervene on the data to explain it. **[TODO: Verify each statement above against the full text of the cited papers.]**

B. Shortcut learning and dataset artifacts

Shortcut learning in deep networks is reviewed in [4]. In language tasks, annotation artifacts, hypothesis-only baselines and syntactic heuristics show that models can score well without solving the intended task [5]–[7]. For fake news, an audit of ISOT/Kaggle reports several leakage channels and a classifier on metadata alone that reaches F1 of 1.0, together with weak transfer to a claim-level target [8]; other work studies length confounds and cross-dataset transfer [11], shortcut measurement and mitigation [9], a replication-based systematization [10], dataset-quality tooling [18], entity bias under temporal shift [12], name masking against diachronic bias [19], and removing opinion articles before transfer experiments [20]. **[TODO: Fill Table I after reading these papers; delete any claim you cannot support from the text.]**

C. Duplicates and label quality

Duplicated training data inflate scores and leak across splits [21], and label errors destabilize benchmarks [22]. Resemblance and containment of documents can be estimated with MinHash-based sketches [23]. We use these ideas to audit the corpora before any modelling.

D. Positioning

Our study differs from the above in combining, under one protocol, (a) cross-dataset gap as the outcome, (b) three artifact families as separately and jointly applied interventions, (c) random-deletion controls matched in size per document, (d)

TABLE I
CLOSEST PRIOR WORK (TO BE COMPLETED FROM THE FULL TEXTS).

Paper	Datasets	Artifact(s)	Cross-data	Gap attribution
[8]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
[11]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
[3]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
[9]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
[12]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
This work	3	S, L, E	yes	yes

overlap- and duplicate-controlled datasets, and (e) per-pair reporting. **[TODO: Confirm with the full-text check that no cited paper already does this combination.]**

V. RESEARCH METHODOLOGY

A. Datasets

We use three English full-text news corpora: ISOT [24] (real articles from Reuters, fake from sites flagged by fact-checkers; 2016–2017), the McIntire “Fake or Real News” set [25], and WELFake [26], a collection merged from several sources. Each document is represented as title plus body. ISOT’s `subject` and `date` fields are never used. LIAR [27] and FakeNewsNet [28] were considered but excluded (short claims; crawling required).

B. Audit and cleaning

Normalization. Two passes: lowercase, punctuation and whitespace collapse; and additionally removal of URLs, handles, a leading “CITY (Reuters) –” dateline, agency tags and the word “Reuters”, so that the same story cleaned differently still matches (used for matching only). *Matching.* Exact hashes (128-bit) of text and of title+text; a 300-character prefix match; and near-duplicates by MinHash (128 permutations, 32 bands $\times$ 4 rows) over word 5-shingles on the full text and on the first 100 words, with candidates re-checked by the true Jaccard similarity (threshold 0.8) or containment (≥ 0.9). A sensitivity tier uses 3-shingles with Jaccard ≥ 0.5 . *Cleaning.* Documents with fewer than 20 words and exact duplicates are removed within each dataset; near-duplicate clusters (Jaccard ≥ 0.8) are kept but recorded, and the later splits are group-aware on these clusters. *Label direction.* Documentation of WELFake is contradictory, so we verified it on Reuters-tagged articles (21,266 under label 0, 16 under label 1): label 0 is real and label 1 is fake. After this correction, exact-text label conflicts between datasets fell to one per pair.

C. Analysis set and pre-registered thresholds

Because WELFake contains all cleaned ISOT and McIntire documents (Section VI-A), we remove from WELFake every

document that is an exact, prefix or near-duplicate match of an ISOT or McIntire document, leaving a *WELFake residual*. Before seeing the overlap, we set the rule: keep a dataset if every class has at least 3,000 documents and the minority class is at least 40% of the dataset. The rule narrowly failed (McIntire real: 2,964; residual minority share: 39.2%). We proceeded and report this as a deviation; Section VI-E shows a duplicate-removal rerun.

D. Interventions

Each intervention deletes spans from the text and is applied to training and test documents alike.

- **S (source markers):** regular expressions for URLs and broken URLs, datelines and agency tags, outlet names, image credits, “share/tweet/print” prompts, advertisement markers, posted-at stamps, “Source:” credit lines, email addresses, embedded-tweet attributions, and @/# handles. The patterns were written after inspecting label-predictive tokens, and S therefore removes *overt* markers, not all source information. S is split post hoc into W (datelines, agency tags, outlet names) and B (all other web boilerplate).
- **L (length matching):** within each dataset, documents are binned into ten deciles of word count, and the same number of real and fake documents is sampled per bin, so length carries no class information within a dataset.
- **E (entities):** spans labelled PERSON, ORG, GPE, NORP, LOC, FAC, EVENT, PRODUCT, WORK_OF_ART, LAW or LANGUAGE by spaCy’s `en_core_web_sm` [29] are deleted.
- **T (temporal expressions, exploratory):** weekdays, capitalized month names with optional day and year, four-digit years, numeric dates and clock times. T was added after inspecting residual cues.
- **Random controls** R_S , R_E , R_{SE} (R_W , R_B , R_T for the exploratory parts): for every document, delete the same number of whole words as the real intervention removed, at random positions.

The planned design is the 2^3 factorial over S, L, E (eight cells) with the three random controls; T and the S split are exploratory.

E. Experimental protocol

Universe. Up to 4,000 documents per dataset and class are drawn once (22,005 documents). *Splits.* For each seed, 20% of near-duplicate clusters of each dataset are held out as the in-domain test pool; the cross-dataset test sample is drawn from all documents of the target. *Sampling.* For each dataset and seed, the training, in-domain and cross-dataset sample sizes are fixed (about 1,495 per class for training, 400–500 per class in-domain, about 795 per class cross-dataset) and *identical across all conditions*, with exactly 50% fake, so conditions differ only in the intervention. *Seeds.* Five seeds change the split and the sample.

TABLE II
DATASETS. RAW AND CLEANED SIZES, AND THE ANALYSIS SET (REAL/FAKE).

Dataset	Raw	Clean	Analysis (R/F)	Words (R/F)
ISOT	44,898	38,030	20,923/17,107	395/448
McIntire	6,335	6,005	2,964/3,041	939/709
WELFake (resid.)	72,134	61,734	10,427/6,735	895/640

Cleaning removed short/empty documents (1,106; 211; 1,799) and exact duplicates (5,762; 119; 8,601). The residual excludes 44,572 WELFake documents that overlap ISOT or McIntire. Mean words are for title plus body.

F. Models, metrics and statistics

TF-IDF + LR: word 1–2-grams, minimum document frequency 2, maximum 0.95, sublinear term frequency, at most 200,000 features, logistic regression ($C=1$, liblinear) [30]. *DistilBERT* [31], a distilled version of BERT [32], with the Transformers library [33]: `distilbert-base-uncased`, first 256 tokens, 2 epochs, batch 16, AdamW with learning rate 3×10^{-5} , weight decay 0.01, 6% warm-up, mixed precision on a GPU **[TODO: state the GPU]**. It reuses the exact samples of the TF-IDF arm.

Metrics. ROC-AUC is the headline metric. F1 of the fake class is also reported but is less informative: with balanced classes, predicting “fake” for everything gives F1 of 0.667 and AUC of 0.5.

Statistics. The six ordered pairs within a seed share training sets, so they are not independent. We average pairs within each seed and report the mean with a 95% *t*-interval over the five seed means (four degrees of freedom) and the number of seeds whose effect has the same sign. These intervals describe resampling variation for these three datasets, not uncertainty about other datasets. Factorial effects use ± 1 coding: an effect is the average change when a factor (or interaction) goes from -1 to $+1$. Nothing is corrected for multiple comparisons, and conditions beyond the factorial are labelled exploratory.

VI. RESULTS AND DISCUSSION

A. Audit results

Table II summarizes the cleaning. ISOT lost 5,762 exact duplicates (12.8% of its 44,898 rows) and WELFake 8,601 (11.9%). All 38,030 cleaned ISOT documents and all 6,005 cleaned McIntire documents have an exact-text match in WELFake, whereas ISOT and McIntire share only 2 exact and about 33 near-duplicate documents (0.1% and 0.5%). WELFake is therefore not an independent third dataset; its residual (17,162 documents, 10,427 real and 6,735 fake) is.

B. TF-IDF: pooled results

Table III and Fig. 1 give the pooled results. The baseline AUC gap is 0.212. Applying all three interventions gives 0.157, which removes 25.8% of the gap (95% interval 21.4–30.3%); length matching and entity masking together remove 26.8% (22.2–31.5%), and source masking adds nothing. Random-deletion controls leave the gap unchanged (0.211–0.224). The factorial effects on the AUC gap (Table IV) are -0.037 for L and -0.023 for E, with interactions of at

TABLE III

TF-IDF + LR, POOLED OVER THE SIX ORDERED PAIRS, FIVE SEEDS.
IN-DOMAIN AUC, CROSS-DATASET AUC AND AUC GAP WITH 95%
INTERVALS OVER SEEDS.

Condition	In AUC	Cross AUC	Gap
C0 (baseline)	0.986	0.774 [.770, .777]	0.212 [.207, .217]
S	0.978	0.759 [.751, .766]	0.219 [.213, .225]
L	0.985	0.805 [.800, .810]	0.180 [.174, .186]
E	0.979	0.787 [.784, .790]	0.191 [.189, .194]
S+L	0.978	0.801 [.794, .807]	0.178 [.171, .184]
S+E	0.972	0.777 [.768, .786]	0.195 [.186, .204]
L+E	0.981	0.826 [.818, .833]	0.155 [.147, .164]
S+L+E	0.973	0.816 [.811, .820]	0.157 [.149, .166]
R_S	0.985	0.775 [.771, .779]	0.211 [.207, .215]
R_E	0.984	0.766 [.762, .770]	0.218 [.215, .220]
R_{SE}	0.984	0.759 [.756, .763]	0.224 [.221, .228]

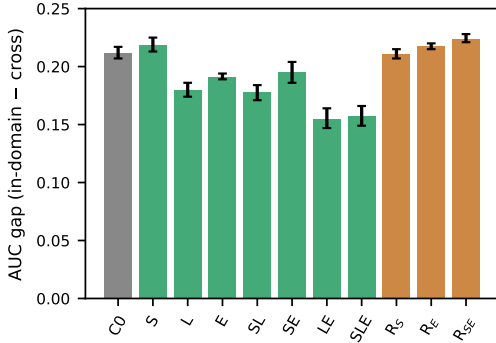


Fig. 1. Pooled AUC gap by condition for TF-IDF + LR (mean, 95% interval over five seeds). Gray: baseline; green: factorial cells; orange: random-deletion controls.

most 0.003 in magnitude, so the effects are close to additive. Length matching raises cross-dataset AUC from 0.774 to 0.805 while leaving in-domain AUC unchanged (0.986 to 0.985): the gap shrinks because transfer improves, not because in-domain performance falls.

Against matched random deletion, entity masking raises cross-dataset AUC by 0.021 [0.017, 0.025] and lowers the gap by 0.026 [0.029, 0.024]. Source masking has the opposite effect: cross-dataset AUC changes by -0.016 [-0.023 , -0.008] and the gap grows by 0.009 [0.001, 0.016].

C. Effects differ by dataset pair

Table V and Fig. 2 show that the pooled picture hides large differences between pairs. Length matching and entity masking raise cross-dataset AUC for all four transfers involving ISOT (L+E: $+0.040$ to $+0.135$). For the two transfers between McIntire and the residual, which already start from high AUC (0.915–0.923), L+E *lowers* AUC slightly (-0.018 and -0.016 ; both intervals exclude zero). This matches the direction of the length confound: in ISOT the fake articles are longer (448 vs. 395 words on average), whereas in McIntire and the residual the real ones are (939 vs. 709; 895 vs. 640). A length cue learned on ISOT misleads on the other two datasets, whereas a cue learned on McIntire transfers to the residual because

TABLE IV

TF-IDF FACTORIAL EFFECTS (CODED ± 1) ON CROSS-DATASET AUC AND ON THE AUC GAP, 95% INTERVALS OVER FIVE SEEDS.

Term	Cross AUC	AUC gap
S	-0.010 [-0.013 , -0.007]	$+0.002$ [-0.001 , $.006$]
L	$+0.038$ [$.035$, $.040$]	-0.037 [-0.039 , $-.034$]
E	$+0.017$ [$.011$, $.023$]	-0.023 [-0.029 , $-.016$]
S $\times$ L	$+0.003$ [-0.000 , $.006$]	-0.003 [-0.005 , $.000$]
S $\times$ E	-0.000 [-0.004 , $.004$]	$+0.000$ [-0.003 , $.003$]
L $\times$ E	$+0.001$ [-0.002 , $.004$]	$+0.000$ [-0.003 , $.003$]
S $\times$ L $\times$ E	-0.003 [-0.007 , $.002$]	$+0.002$ [-0.002 , $.006$]

they share its direction. The pooled gain is therefore driven by ISOT-involving pairs. ISOT-trained models remain weak even after masking (cross AUC 0.67–0.69 for S+L+E), so most of the unexplained gap lies in transfers from ISOT.

D. Exploratory analyses: source split and temporal expressions

Table VI reports two post-hoc analyses. Splitting S shows that the harm comes from the web-boilerplate part (share prompts, advertisements, credits, embeds, URLs): its removal lowers cross-dataset AUC by 0.014 [0.022, 0.007] and widens the gap by 0.009, whereas removing wire/outlet markers is neutral. A plausible reading is that such boilerplate is shared across low-quality sites and therefore transfers; it is an artifact for content-based judgement but a genuine signal of publisher type. Removing temporal expressions does not help either: cross-dataset AUC falls by 0.012 and S+L+E+T is 0.016 below S+L+E. We had expected a collection-window artifact to drive McIntire–residual transfer; the data do not support that.

E. Robustness to residual duplicates

We flagged 207 of 61,197 documents in the analysis set that were exact, prefix or near-duplicates of a document in another dataset, or that were near-empty after stripping source markers (ISOT 63, residual 144; McIntire none, by construction). Of these, 94 of 22,005 fell in the sampling universe. Rerunning the baseline, L, E, L+E and the random controls with the same five seeds changed no pair-level sign. The largest change in a point estimate was 0.010, and the pooled share removed by L+E was 28.0% [23.8, 32.1] (26.8% before). One claim weakened: the length effect for McIntire→residual moved from -0.014 to -0.007 with an interval including zero, while the L+E effect stayed at -0.018 .

F. DistilBERT

DistilBERT has a higher baseline cross-dataset AUC (0.804 vs. 0.774) and a smaller gap (0.193 vs. 0.212), but the intervention effects are largely absent once random controls are considered (Fig. 3). Source masking raises cross-dataset AUC from 0.804 to 0.835, but random deletion of the same number of words gives 0.834, and the specific effect of source masking is $+0.000$ [-0.021 , 0.022]. Entity masking is *worse* than random deletion by 0.022 [0.015, 0.028]. Pooled over pairs, random deletion alone moves the AUC gap by about

TABLE V

TF-IDF + LR: CHANGE IN CROSS-DATASET AUC RELATIVE TO THE BASELINE, PER ORDERED PAIR (MEAN OVER FIVE SEEDS; **BOLD**: 95% INTERVAL OVER SEEDS EXCLUDES ZERO). THE FIRST COLUMN IS THE BASELINE AUC.

Pair (train→test)	C0 AUC	S	L	E	L+E	S+L+E	R_E
ISOT→McIntire	.623	-.017	+.053	+.020	+.074	+.066	+.006
ISOT→residual	.632	-.011	+.032	-.003	+.040	+.039	-.004
McIntire→ISOT	.746	-.041	+.058	+.046	+.135	+.113	-.022
McIntire→residual	.915	-.010	-.014	-.006	-.018	-.023	+.001
residual→ISOT	.803	-.005	+.077	+.022	+.096	+.091	-.021
residual→McIntire	.923	-.006	-.018	+.002	-.016	-.033	-.006

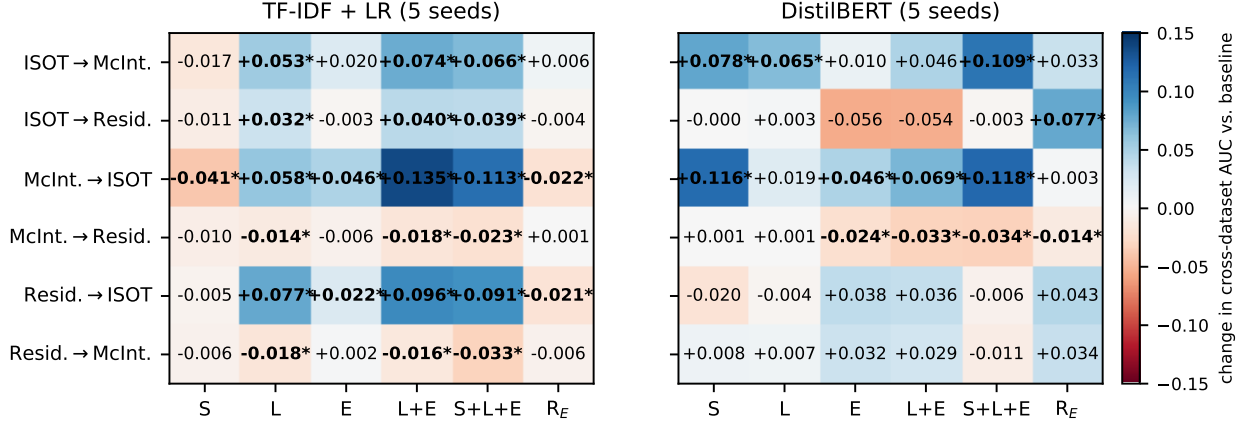


Fig. 2. Change in cross-dataset AUC relative to the baseline for each ordered pair and intervention (TF-IDF + LR left, DistilBERT right). Asterisks and bold: 95% interval over five seeds excludes zero. R_E is the random-deletion control matched to entity masking.

TABLE VI

EXPLORATORY TF-IDF ANALYSES: CHANGE IN CROSS-DATASET AUC VS. THE BASELINE AND VS. A RANDOM-DELETION CONTROL OF THE SAME SIZE, AND CHANGE IN AUC GAP VS. BASELINE (95% INTERVALS OVER SEEDS).

Intervention	$\Delta\text{AUC vs C0}$	$\Delta\text{AUC vs random}$	$\Delta\text{gap vs C0}$
S_W (wire/outlet)	-.002 [-.004, .000]	+.006 [.004, .009]	-.002 [-.004, -.001]
S_B (web boilerplate)	-.014 [-.022, -.007]	-.012 [-.023, .000]	+.009 [.003, .014]
T (temporal)	-.012 [-.019, -.006]	-.009 [-.020, .003]	+.009 [.001, .017]

the same amount as S, E or S+L+E (shares removed: R_S and R_E 15.5%, S+L+E 16.9%), so token dropout acts as a regularizer, an effect that TF-IDF does not show. At pair level, the combined intervention helps ISOT→McIntire (+0.109) and McIntire→ISOT (+0.118) and hurts McIntire→residual (-0.034), as for TF-IDF; the other three pairs are inconclusive (Fig. 2). Run-to-run variation is large: with three seeds, the S+L+E estimate for residual→McIntire was -0.047 with an interval excluding zero, and with five seeds it is -0.011 [-0.115, 0.094]. DistilBERT reads only the first 256 tokens, so document length is largely invisible to it, consistent with the absence of a length effect.

G. Discussion

Artifacts explain a minority of the gap. Even in the model where they matter most, the three families together explain about a quarter. Because our removal is imperfect (regular expressions tuned on these data, a small NER model), this is a lower bound for these families; the remainder cannot

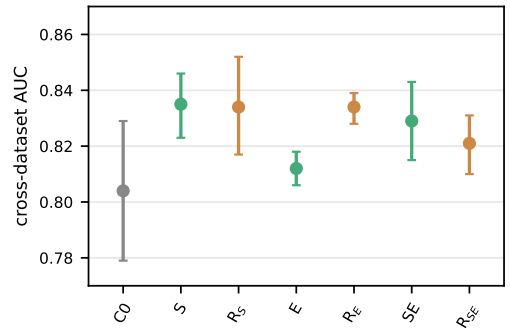


Fig. 3. DistilBERT cross-dataset AUC (mean, 95% interval over five seeds): baseline (gray), real masking (green) and size-matched random deletion (orange). Real masking is not better than random deletion.

be attributed to “genuine” domain shift, only to what our interventions did not remove.

“Artifact” depends on the goal. Web-boilerplate markers are shortcuts if the aim is to judge content, but they are legitimate evidence of publisher type and they transfer across low-quality sites. Removing them lowers transfer.

Random controls matter. For DistilBERT, omitting the random-deletion control would have shown that source masking “improves generalization” by 0.031 AUC; random deletion achieves the same.

Recommendations for dataset-transfer studies: audit overlap and duplicate content before pairing datasets, verify label direction on known content, equalize sample sizes and class balance across conditions, include size-matched random controls, report pairs separately, and use AUC when classes are balanced.

H. Limitations

Three English datasets from 2016–2017 US political news, one of them derived from another (the residual), and labels that may reflect source rather than veracity; two model families, with five seeds each and large run-to-run variation for DistilBERT; source patterns tuned on the data and exploratory conditions (T, S split) added after inspecting results, with no multiple-comparison correction; a pre-registered size rule that was narrowly missed; subsampled training sets (about 1,495 documents per class) and length matching that changes the length range; intervals that describe resampling noise for these datasets only; and a DistilBERT arm that sees only 256 tokens. **[TODO: Add any limitation your instructor or reviewers would raise.]**

VII. CONCLUSION

Across three audited news corpora, removing source markers, matching length and masking entities together explains about a quarter of the cross-dataset AUC gap of a TF-IDF detector, and about three quarters remains. Length matching and entity masking contribute, source masking does not, and the effects depend strongly on the dataset pair. For DistilBERT the same interventions do no better than deleting random words. These results argue for overlap-aware dataset pairing, random-deletion controls and per-pair reporting in transfer studies, and against reading a shrinking gap as proof that an artifact was the cause.

VIII. FUTURE SCOPE

Add datasets with different construction, languages and years; include models such as RoBERTa or instruction-tuned LLMs; replace regular-expression source masking with learned or human-validated marker detection; separate publisher-type signal from veracity signal; test causal interventions such as counterfactual rewriting; and evaluate with human-verified labels and temporally disjoint splits.

REPRODUCIBILITY AND DATA AVAILABILITY

Code for the audit, interventions, experiments and analysis: **[TODO: repository URL]**. We do not redistribute the datasets; the document identifiers of the flagged duplicates,

configuration files and seeds are provided. Software versions: **[TODO: Python, scikit-learn, spaCy and model version, transformers, PyTorch]**.

ACKNOWLEDGMENT

[TODO: Supervisor, institution.] AI assistance: an AI system (Claude, Anthropic) was used for planning the experiments, writing and testing analysis code, and drafting text. The author ran all experiments and verified the results. **[TODO: Check your institution’s AI-use policy and edit this statement accordingly.]**

REFERENCES

- [1] M. Janicka, M. Pszona, and A. Wawer, “Cross-domain failures of fake news detection,” *Computación y Sistemas*, vol. 23, no. 3, 2019, vERIFY.
- [2] N. Hoy and T. Koulouri, “Exploring the generalisability of fake news detection models,” in *Proc. IEEE International Conference on Big Data*, 2022, vERIFY.
- [3] Dell’Oglio *et al.*, “An experimental comparison of the most popular approaches to fake news detection,” arXiv:2603.25501, 2026, vERIFY authors/year/title.
- [4] R. Geirhos *et al.*, “Shortcut learning in deep neural networks,” *Nature Machine Intelligence*, vol. 2, pp. 665–673, 2020, vERIFY.
- [5] S. Gururangan, S. Swayamdipta, O. Levy, R. Schwartz, S. Bowman, and N. A. Smith, “Annotation artifacts in natural language inference data,” in *Proc. NAACL-HLT, Short Papers*, 2018, pp. 107–112, vERIFY.
- [6] A. Poliak, J. Naradowsky, A. Haldar, R. Rudinger, and B. Van Durme, “Hypothesis only baselines in natural language inference,” in *Proc. 7th Joint Conference on Lexical and Computational Semantics (*SEM)*, 2018, pp. 180–191, vERIFY.
- [7] T. McCoy, E. Pavlick, and T. Linzen, “Right for the wrong reasons: Diagnosing syntactic heuristics in natural language inference,” in *Proc. 57th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2019, pp. 3428–3448, vERIFY.
- [8] vERIFY AUTHORS, “What does 99% accuracy measure? a reproducible audit of shortcut learning in a widely used fake news corpus,” arXiv:2609.25006, 2026, vERIFY authors/year; read the full text before citing specific numbers.
- [9] Wan *et al.*, “Truth over tricks: Measuring and mitigating shortcut learning in misinformation detection,” arXiv:2506.02350, 2025, vERIFY authors/venue.
- [10] A. Xiao and J. Mayer, “SoK: Machine learning for misinformation detection,” arXiv:2308.12215, 2023, vERIFY authors/venue (USENIX Security 2025 per the literature report).
- [11] vERIFY AUTHORS, “When benchmarks mislead: Shortcut learning, length confounds, and the limits of cross-dataset generalization in multilingual fake news and sarcasm detection,” arXiv:2607.14131, 2026, vERIFY authors/year; check how WELFake labels were mapped before discussing their WELFake-to-ISOT result.
- [12] Y. Zhu *et al.*, “Generalizing to the future: Mitigating entity bias in fake news detection,” in *Proc. 45th International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2022, vERIFY authors; DOI 10.1145/3477495.3531816; arXiv 2204.09484.
- [13] K. Shu, A. Sliva, S. Wang, J. Tang, and H. Liu, “Fake news detection on social media: A data mining perspective,” *ACM SIGKDD Explorations Newsletter*, vol. 19, no. 1, pp. 22–36, 2017, vERIFY.
- [14] X. Zhou and R. Zafarani, “A survey of fake news: Fundamental theories, detection methods, and opportunities,” *ACM Computing Surveys*, vol. 53, no. 5, pp. 1–40, 2020, vERIFY.
- [15] D. M. J. Lazer *et al.*, “The science of fake news,” *Science*, vol. 359, no. 6380, pp. 1094–1096, 2018, vERIFY full author list.
- [16] L. Bozarth and C. Budak, “Toward a better performance evaluation framework for fake news classification,” in *Proc. International AAAI Conference on Web and Social Media (ICWSM)*, vol. 14, 2020, vERIFY pages; DOI 10.1609/icwsml.v14i1.7279 per the literature report.
- [17] F. T. Asr and M. Taboada, “Big data and quality data for fake news and misinformation detection,” *Big Data & Society*, vol. 6, no. 1, 2019, vERIFY.

- [18] Thibault *et al.*, “A guide to misinformation detection data and evaluation,” arXiv:2411.05060, 2024, vERIFY authors/venue (KDD 2025 per the literature report). Only the abstract was read.
- [19] Murayama *et al.*, “Mitigation of diachronic bias in fake news detection dataset,” arXiv:2108.12601, 2021, vERIFY authors/title/venue. Only the abstract was read.
- [20] C. Blackledge and A. Atapour-Abarghouei, “Transforming fake news: Robust generalisable news classification using transformers,” arXiv:2109.09796, 2021, vERIFY.
- [21] K. Lee *et al.*, “Deduplicating training data makes language models better,” in *Proc. 60th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2022, vERIFY authors/pages.
- [22] C. G. Northcutt, A. Athalye, and J. Mueller, “Pervasive label errors in test sets destabilize machine learning benchmarks,” in *NeurIPS Datasets and Benchmarks Track*, 2021, vERIFY.
- [23] A. Z. Broder, “On the resemblance and containment of documents,” in *Proc. Compression and Complexity of Sequences*, 1997, pp. 21–29, vERIFY.
- [24] H. Ahmed, I. Traore, and S. Saad, “Detection of online fake news using N-gram analysis and machine learning techniques,” in *Intelligent, Secure, and Dependable Systems in Distributed and Cloud Environments (ISDDC)*, ser. LNCS, vol. 10618, 2017, pp. 127–138, vERIFY pages.
- [25] G. McIntire, “Fake or real news (McIntire dataset),” Kaggle / GitHub dataset, 2017, vERIFY: cite the exact source you downloaded it from.
- [26] P. K. Verma, P. Agrawal, I. Amorim, and R. Prodan, “WELFake: Word embedding over linguistic features for fake news detection,” *IEEE Transactions on Computational Social Systems*, vol. 8, no. 4, pp. 881–893, 2021, vERIFY.
- [27] W. Y. Wang, ““Liar, liar pants on fire”: A new benchmark dataset for fake news detection,” in *Proc. 55th Annual Meeting of the Association for Computational Linguistics (ACL), Short Papers*, 2017, pp. 422–426, vERIFY.
- [28] K. Shu, D. Mahudeswaran, S. Wang, D. Lee, and H. Liu, “FakeNewsNet: A data repository with news content, social context, and spatiotemporal information for studying fake news on social media,” *Big Data*, vol. 8, no. 3, pp. 171–188, 2020, vERIFY.
- [29] M. Honnibal, I. Montani, S. Van Landeghem, and A. Boyd, “spaCy: Industrial-strength natural language processing in Python,” Software, <https://spacy.io>, 2020, vERIFY the preferred citation; state the model name (en_core_web_sm) and version used.
- [30] F. Pedregosa *et al.*, “Scikit-learn: Machine learning in Python,” *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011, vERIFY.
- [31] V. Sanh, L. Debut, J. Chaumond, and T. Wolf, “DistilBERT, a distilled version of BERT: Smaller, faster, cheaper and lighter,” arXiv:1910.01108, 2019, vERIFY.
- [32] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, “BERT: Pre-training of deep bidirectional transformers for language understanding,” in *Proc. NAACL-HLT*, 2019, pp. 4171–4186, vERIFY.
- [33] T. Wolf *et al.*, “Transformers: State-of-the-art natural language processing,” in *Proc. EMNLP: System Demonstrations*, 2020, pp. 38–45, vERIFY.